

Composite Operators in the Perturbation Theory of Renormalizable Interactions^{*†}

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In perturbation theory the Green's functions of composite operators are constructed by applying Bogoliubov's method of renormalization. Normalization conditions for the composite operators are derived, as well as identities which relate composite operators of different degree.

1. INTRODUCTION

The purpose of this paper is to define composite operators and discuss some of their properties in the perturbation theory of renormalizable interactions. As an example we consider the model of a scalar field with a mixed A^3 - and A^4 -coupling

$$\mathcal{L}\{A\} = \mathcal{L}_0\{A\} + \mathcal{L}_{\text{int}}\{A\} \quad (1.1)$$

$$\mathcal{L}_0\{A\} = (1/2) \partial_\mu A \partial^\mu A - (m^2/2) A^2 \quad (1.2)$$

$$\mathcal{L}_{\text{int}}\{A\} = -\frac{g}{3!} A^3 - \frac{\lambda}{4!} A^4 + \frac{a}{2} A^2 + b \mathcal{L}_0\{A\} \quad (1.3)$$

$$\lambda \neq 0 \quad (1.4)$$

$\mathcal{L}\{A\}$ will be used as an effective Lagrangian in the following sense. m , g and λ represent the renormalized mass and coupling constants. The parameters a and b are power series in g and λ with finite coefficients depending on m . The coefficients are recursively defined by the condition that the propagator has a pole at m^2 of residue i . In contradistinction to the Lagrangian of conventional renormalization theory $\mathcal{L}\{A\}$ does not contain infinite parameters.

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The time ordered Green's functions of the theory are defined by the Gell-Mann Low expansion [1]

$$\langle TA(x_1) \cdots A(x_n) \rangle = \text{Fin}(\langle \text{Te}^{i \int_{\text{int}}^0 dz A^0(x_1) \cdots A^0(x_n)} \rangle_0 / \langle \text{Te}^{i \int_{\text{int}}^0 dz} \rangle_0) \quad (1.5)$$

$$\mathcal{L}_{\text{int}}^0 = \mathcal{L}_{\text{int}}\{A^0(z)\}$$

A^0 denotes a free scalar field of mass m , $\langle \rangle_0$ the free field vacuum expectation value. The symbol 'Fin' indicates that the finite part of the contribution from each Feynman diagram should be taken. Various methods for separating the finite part from a divergent Feynman integral are available [2–7]¹. In the original version of Bogoliubov's method the finite part is first defined for the regularized contribution from a given diagram [2]. As was shown by Hepp the finite part thus defined approaches a well defined distribution when the regularization is removed [3]. In this paper Bogoliubov's method is used in the version of [5]. Without regularization the finite part is separated by taking suitable subtractions for the integrand of a Feynman integral in momentum space, as in the original work of Dyson and Salam [8–9]. The appropriate subtraction terms are obtained by applying Bogoliubov's R -operation to the unrenormalized integrand. The resulting integrals can be shown to be convergent as a consequence of the power counting theorem [5].

Composite operators $B(x)$ will be defined through their Green's function

$$\langle TB(x) A(y_1) \cdots A(y_n) \rangle \quad (1.6)$$

The matrix elements of $B(x)$ between incoming and outgoing states can be expressed in terms of the Green's functions (1.6) by using the reduction formulas. It is not obvious, however, that the time ordered Green's functions of the operator thus defined are identical to the original functions (1.6). For this certain consistency conditions² on the corresponding retarded functions are required which also yield the locality of B . This point will not be discussed in the present paper.

Let

$$M = M\{A(x)\} \quad (1.7)$$

be a monomial of the field A and its derivatives. Our aim is to define an operator B which can be interpreted as the finite part of (1.7). Formally the time ordered Green's functions of M are given by the Gell-Mann Low expansion

$$\langle TM\{A(x)\} A(y_1) \cdots A(y_n) \rangle = (\langle \text{Te}^{i \int_{\text{int}}^0 dz M\{A^0(x)\} A^0(y_1) \cdots A^0(y_n)} \rangle_0 / \langle \text{Te}^{i \int_{\text{int}}^0 dz} \rangle_0) \quad (1.8)$$

This suggests defining the Green's functions

$$\langle TB(x) A(y_1) \cdots A(y_n) \rangle \quad (1.9)$$

¹ The Epstein-Glaser method [6] including some applications is reviewed in ref. [7].

² For a formulation of the consistency conditions see [10].

of the composite operator B by taking the finite part of (1.8). B may be called a normal product of (1.7), in generalization of Wick's definition of a normal product in the free field case [11].

The Feynman diagrams appearing in the expansion (1.8) contain a new vertex x which requires additional subtractions. Depending on the number of subtractions employed for removing the new divergencies we obtain different versions of the normal product. The minimal normal product of (1.7) is constructed by making only those subtractions which are required for removing all divergencies. In this paper a family

$$N_a[M\{A(x)\}] \quad (1.10)$$

of normal products will be introduced. The normal product (1.10) of degree a is constructed by making $a - r + 1$ over-all subtractions for any proper diagram which contains x and has r external lines. It will be shown that this operator is finite provided

$$a \geq d(M) \quad (1.11)$$

where $d(M)$ is the dimension of the monomial M . If the degree equals the dimension we use the notation

$$\begin{aligned} N[M\{A(x)\}] &= N_a[M\{A(x)\}], \\ a &= d(M), \end{aligned} \quad (1.12)$$

and say that the normal product is of minimal degree. If the A^3 -coupling is absent, i.e., $g = 0$, the operators (1.12) represent the minimal normal products. For $g \neq 0$ the operators (1.12) are not minimal. In the general case the minimal operators do not seem to be very useful and will not be considered in this paper.

Alternative methods of defining composite operators have been developed by Epstein-Glaser [6, 7] and Wilson [12]. Wilson's method uses a renormalized version of the relations

$$\begin{aligned} \frac{\partial A(x)}{\partial \lambda} &= -i \int_{-\infty}^{x_0} dy_0 \int d\mathbf{y} [\mathcal{L}_{\text{int}}(y), A(x)] \\ \frac{\partial \mathcal{L}_{\text{int}}(x)}{\partial \lambda} &= -i \int_{-\infty}^{x_0} dy_0 \int d\mathbf{y} [\mathcal{L}_{\text{int}}(y), \mathcal{L}_{\text{int}}(x)] \end{aligned} \quad (1.13)$$

On the basis of these relations the short distance expansion and composite operators are constructed recursively in perturbation theory.

In the method of Epstein and Glaser composite operators are first constructed for a model where the coupling constant is replaced by an external source $\lambda(x)$.

The adiabatic limit $\lambda(x) \rightarrow \lambda$ is then shown to exist [6, 7]. One advantage is that composite operators are constructed directly as local operators—without reference to Green's functions. On the other hand the method described in the present paper seems to be more suitable for investigating anomalies [18, 19, 26].

In Section 2 some definitions of Green's functions are collected. After a review of the renormalization of Feynman integrals (Section 3) the Green's functions of composite operators are constructed and their normalization conditions derived (Section 4). The integrands of Feynman integrals contributing to composite operators satisfy certain algebraic identities which will be derived in Section 5. These identities imply that any normal product

$$N_b[M] \quad (1.14)$$

may be written as a linear combination of normal products of degree a , for any given integer $a > b$

$$N_b[M] = \sum c_{ba}(M, M') N_a[M'] \quad (1.15)$$

(Section 6). The sum contains only a finite number of terms, it extends over all monomials M' for which the dimension $d(M')$ is less or equal than the degree a ,

$$d(M') \leq a. \quad (1.16)$$

As a consequence of (1.15) any composite operator (1.10) may be linearly expressed in terms of the operators (1.12).

We finally review some applications of the normal product algorithm [17–19, 24–29]. The field operator A satisfies a finite field equation which has the form of the Euler–Lagrange equation derived from the effective Lagrangian (1.1–3), but with normal products N_3 applied to the interaction terms. The Wilson expansion may be used to express the normal product in terms of A by the point-splitting method. In this way Valatin's form of the field equation is obtained [13–17]³.

Lowenstein [18] constructed a finite energy–momentum tensor [19–22]⁴

$$T_{\mu\nu} = N_4[\theta_{\mu\nu}] + c(\partial_\mu \partial_\nu - g_{\mu\nu} \square) N_2[A^2] \quad (1.17)$$

in perturbation theory which has all required properties, in particular the appropriate Ward identities are satisfied. $\theta_{\mu\nu}$ is the formal energy–momentum tensor

$$\theta_{\mu\nu} = (\partial \mathcal{L} / \partial \partial^\mu A) \partial_\nu A - g_{\mu\nu} \mathcal{L}$$

of the Lagrangian (1.1–3), c is an arbitrary constant. As was shown by Lowenstein

³ A derivation of local field equations in the framework of the Epstein-Glaser method was given by Stora [7].

⁴ The improved energy-momentum tensor introduced in [21] for a regularized theory with A^4 -coupling does not have a finite limit when the regularization is removed. Symanzik [22] found a corrected form of this tensor containing one free parameter which in the limit approaches (1.17).

the trace of the tensor (1.17) is never soft in perturbation theory, no matter how c is chosen.

In a paper by Schroer it was shown that independent of perturbation theory the trace of the tensor (1.17) is soft for suitable c provided a Gell-Mann Low eigenvalue of the coupling constant exists. This would imply the asymptotic conformal invariance of the Green's functions [23].

Lowenstein [25] found an elegant proof of the Callan–Symanzik equation [23] and derived a differential equation of the renormalization group. Anomalous Ward identities and gauge invariance problems were studied for various cases [26]. Field equations were derived for a very general class of interactions [27, 29]. Lam proposed a generalization of the subtraction scheme which is particularly useful for studying nonlinear transformations [27]. Finally, the normal product algorithm has been extended by Lowenstein to Green's functions involving several composite operators [25].

2. NOTATION OF GREEN'S FUNCTIONS

In this section we collect some definitions concerning Green's functions. For the Fourier transform $\tilde{f}$ of a function f of n four vectors $x_1, \dots, x_n$ we use the unsymmetric definition

$$\tilde{f}(p_1 \cdots p_n) = \int dx_1 \cdots dx_n e^{i\sum p_j x_j} f(x_1 \cdots x_n) \quad (2.1)$$

$$f(x_1 \cdots x_n) = \int \frac{dp_1}{(2\pi)^4} \cdots \int \frac{dp_n}{(2\pi)^4} e^{-i\sum p_j x_j} \tilde{f}(p_1 \cdots p_n)$$

Let $TO_1(x_1) \cdots O_n(x_n)$ be a suitably defined time ordered product of components $O_1, \dots, O_n$ of local field operators such as the scalar field A , derivatives thereof or composite field operators. The Fourier transform or partial Fourier transform of $TO_1(x_1) \cdots O_n(x_n)$ will be denoted by

$$T\tilde{O}_1(p_1) \cdots \tilde{O}_n(p_n) = \int dx_1 \cdots dx_n e^{i\sum p_j x_j} TO_1(x_1) \cdots O_n(x_n) \quad (2.2)$$

$$\begin{aligned} TO_1(x_1) \cdots O_r(x_r) \tilde{O}_{r+1}(p_{r+1}) \cdots \tilde{O}_n(p_n) \\ = \int dx_{r+1} \cdots dx_n e^{i\sum_{j=r+1}^n p_j x_j} TO_1(x_1) \cdots O_n(x_n) \end{aligned} \quad (2.3)$$

For the Fourier transform $\tilde{\Delta}_F'(p)$ of the propagator

$$\Delta_F'(z) = \langle TA(x) A(y) \rangle, \quad z = x - y \quad (2.4)$$

we have

$$\langle T\tilde{A}(p_1) \tilde{A}(p_2) \rangle = (2\pi)^4 \delta(p_1 + p_2) \tilde{\Delta}_F'(p_1) \quad (2.5)$$

and

$$\tilde{A}'_F(p) = \langle TA(0) \tilde{A}(p) \rangle \quad (2.6)$$

$\langle \rangle$ denotes the vacuum expectation value.

The connected part of a time ordered vacuum expectation value is recursively defined by

$$\langle TO_1(x_1) \cdots O_n(x_n) \rangle = \sum \langle TO_{i_{11}}(x_{i_{11}}) \cdots \rangle^{\text{conn}} \cdots \langle TO_{i_{A1}}(x_{i_{A1}}) \cdots \rangle^{\text{conn}} \quad (2.7)$$

The sum extends over all partitions of $x_1, \dots, x_n$ into classes

$$(x_{i_{11}}, x_{i_{21}}, \dots), (x_{i_{21}}, x_{i_{22}}, \dots), \dots, (x_{i_{A1}}, x_{i_{A2}}, \dots)$$

The Fourier transform of the connected part is of the form

$$\langle T\tilde{O}_1(p_1) \cdots \tilde{O}_n(p_n) \rangle^{\text{conn}} = (2\pi)^4 \delta\left(\sum p_j\right) \langle T\tilde{O}_1(p_1) \cdots \tilde{O}_n(p_n) \rangle^{\text{trunc}} \quad (2.8)$$

$\langle \rangle^{\text{trunc}}$ is called the truncated part. A Fourier transformation with respect to p_1 yields

$$\langle T_1\tilde{O}(p_1) \cdots \tilde{O}_n(p_n) \rangle^{\text{trunc}} = \langle T\tilde{O}_1(0) \tilde{O}_2(p_2) \cdots \tilde{O}_n(p_n) \rangle^{\text{conn}} \quad (2.9)$$

The generalized Wick product

$$:\tilde{O}_1(x_1) \cdots \tilde{O}_n(x_n):$$

is defined recursively by

$$TO_1(x_1) \cdots O_n(x_n) = :O_1(x_1) \cdots O_n(x_n): + \sum \langle TO_{i_{01}}(x_{i_{01}}) \cdots \rangle :O_{i_{11}}(x_{i_{11}}) \cdots \quad (2.10)$$

The sum extends over all partitions of $(x_1, \dots, x_n)$ into classes

$$(x_{i_{01}}, x_{i_{02}}, \dots), (x_{i_{11}}, x_{i_{12}}, \dots)$$

where the second class may be empty.

The mixed product

$$T:O_1(x_1) \cdots O_n(x_n): O_1'(y_1) \cdots O_m'(y_m)$$

is defined by

$$\begin{aligned} & TO_1(x_1) \cdots O_n(x_n) O_1'(y_1) \cdots O_m'(y_m) \\ &= T:O_1(x_1) \cdots O_n(x_n): O_1'(y_1) \cdots O_m'(y_m) \\ &+ \sum \langle TO_{i_{01}}(x_{i_{01}}) \cdots \rangle T:O_{i_{11}}(x_{i_{11}}) \cdots :O_1'(y_1) \cdots O_m'(y). \end{aligned} \quad (2.11)$$

Next we introduce the factorization by

$$\begin{aligned} & \langle T\{O_1(x_1) \cdots O_n(x_n)\} O'_1(y_1) \cdots O'_m(y_m) \rangle^{\text{fact}} \\ &= \sum \langle TO'_{i_{01}}(y_{i_{01}}) \cdots \rangle \langle TO_1(x_1) O'_{i_{11}}(y_{i_{11}}) \cdots \rangle^{\text{conn}} \times \cdots \\ & \times \langle TO_n(x_n) O'_{i_{n1}}(y_{i_{n1}}) \cdots \rangle^{\text{conn}} \end{aligned} \quad (2.12)$$

The sum extends over all ordered partitions of $y_1, \dots, y_m$ into $n + 1$ classes

$$(y_{i_{01}}, y_{i_{02}}, \dots), (y_{i_{11}}, y_{i_{12}}, \dots), \dots, (y_{i_{n1}}, y_{i_{n2}}, \dots)$$

$(y_{i_{j1}} \cdots)$ is assigned to x_j ($j = 1, \dots, n$). $(y_{i_{01}} \cdots)$ may be empty.

We give some simple properties of the factorization. For $m = n$

$$\langle T\{A(x_1) \cdots A(x_n)\} A(y_1) \cdots A(y_n) \rangle^{\text{fact}} = \sum \Delta'_F(x_{i_1} - y_1) \cdots \Delta'_F(x_{i_n} - y_n) \quad (2.13)$$

with the sum taken over all permutations of the x_j .

$$\begin{aligned} \langle T\{O_1(x_1) \cdots O_n(x_n)\} O'_1(y_1) \cdots O'_m(y_m) \rangle^{\text{fact}} &= 0 \quad \text{for } n > m \\ \langle T\{O(x)\} O_1(y_1) \cdots O_m(y_m) \rangle^{\text{fact}} &= \langle TO(x) O_1(y_1) \cdots O_m(y_m) \rangle \end{aligned} \quad (2.14)$$

Let $G_m(x_1, \dots, x_n; y_1, \dots, y_m)$ be a set of Green's functions which are symmetric in $y_1, \dots, y_m$. We may take, for instance,

$$G_m(x_1 \cdots x_n; y_1 \cdots y_m) = \langle T : A(x_1) \cdots A(x_n) : A(y_1) \cdots A(y_m) \rangle \quad (2.15)$$

or

$$G_m(x; y_1 \cdots y_m) = \langle TB(x) A(y_1) \cdots A(y_m) \rangle \quad (2.16)$$

where $B(x)$ is a composite operator. We introduce the proper part G_m^{prop} of G_m by the recursion formula

$$\begin{aligned} G_m(x_1 \cdots x_n; y_1 \cdots y_m) &= \sum_{a=1}^m \frac{1}{a!} \int dz_1 \cdots dz_a G_a(x_1 \cdots x_n; z_1 \cdots z_a)^{\text{prop}} \\ & \times \langle T\{A(z_1) \cdots A(z_a)\} A(y_1) \cdots A(y_m) \rangle^{\text{fact}} \end{aligned} \quad (2.17)$$

For a unique definition we must require G_m^{prop} to be symmetric in $y_1, \dots, y_m$. We give a few examples. For the set

$$G_m(x; y_1 \cdots y_m) = \langle TA(x) A(y_1) \cdots A(y_m) \rangle$$

of Green's functions we have

$$\begin{aligned} \langle TA(x) A(y) \rangle^{\text{prop}} &= \delta(x - y) \\ \langle TA(x) A(y_1) \cdots A(y_m) \rangle^{\text{prop}} &= 0 \quad \text{if } m > 1. \end{aligned} \quad (2.18)$$

For (2.15) we find

$$\langle T:\tilde{A}(p_1) \cdots \tilde{A}(p_n):\tilde{A}(q)\rangle^{\text{prop}} = \frac{\langle T:\tilde{A}(p_1) \cdots \tilde{A}(p_n):\tilde{A}(q)\rangle}{\tilde{\Delta}_{F'}(p)} \quad (2.19)$$

For later reference we give the free field value of the proper parts

$$\langle T:A_0(x_1) \cdots A_0(x_n):A_0(y_1) \cdots A_0(y_m)\rangle_0^{\text{prop}} = \delta_{mn} \sum_P \delta(x_1 - y_{i_1}) \cdots \delta(x_n - y_{i_n}) \quad (2.20)$$

with the sum extending over all permutations P of $y_1, \dots, y_n$. Derivatives of the field operator will be denoted by

$$A_{(\mu)}(x) = \partial_{(\mu)} A(x) \quad (2.21)$$

where

$$\begin{aligned} (\mu) &= (\mu_1, \dots, \mu_a) \\ \partial_{(\mu)} &= \partial_{\mu_1} \cdots \partial_{\mu_a}. \end{aligned} \quad (2.22)$$

The time ordered product of field operator derivatives is defined by

$$\begin{aligned} TA_{(\mu)_1}(x_1) \cdots A_{(\mu)_n}(x_n) &= \partial_{(\mu)_1}^{x_1} \cdots \partial_{(\mu)_n}^{x_n} TA(x_1) \cdots A(x_n) \\ (\mu)_j &= (\mu_{j1}, \dots, \mu_{jn(j)}) \\ \partial_{(\mu)_j}^{x_j} &= \partial_{\mu_{j1}}^{x_j} \cdots \partial_{\mu_{jn(j)}}^{x_j}, \quad \partial_{\mu_{j\alpha}}^{x_j} = \partial/\partial(x_j)^{\mu_{j\alpha}}. \end{aligned} \quad (2.23)$$

We will also use abbreviations of the form

$$\begin{aligned} T(A_{(\mu)_1}(x_1) \cdots A_{(\mu)_m}(x_m) \tilde{A}^{(n)_1}(q_1) \cdots \tilde{A}^{(n)_\nu}(q_\nu)) \\ = \partial_{(\mu)_1}^{x_1} \cdots \partial_{(\mu)_m}^{x_m} \partial_{q_1}^{(n)_1} \cdots \partial_{q_\nu}^{(n)_\nu} T(A(x_1) \cdots A(x_m) \tilde{A}(q_1) \cdots \tilde{A}(q_\nu)) \end{aligned} \quad (2.24)$$

3. FEYNMAN RULES FOR RENORMALIZED GREEN'S FUNCTIONS

In perturbation theory the Green's function

$$\langle TA(x_1) \cdots A(x_n) \rangle \quad (3.1)$$

is defined by the finite part of the Gell-Mann Low expansion (1.5) which we write as an expansion with respect to Feynman diagrams

$$\langle TA(x_1) \cdots A(x_n) \rangle = \sum_{C \in \mathcal{C}} \langle TA(x_1) \cdots A(x_n) \rangle_C \quad (3.2)$$

The sum extends over the set $\mathcal{C}_n$ of Feynman diagrams with endpoints $x_1, \dots, x_n$ which can be built by using 4-vertices (corresponding to the A^4 -term), 3-vertices (corresponding to the A^3 -term) and 2-vertices (corresponding to the quadratic terms in $\mathcal{L}_{\text{int}}$). Disconnected closed loops (i.e., connected components to which no coordinates x_j are attached) are excluded.

Let $\Gamma \in \mathcal{C}_n$ be a diagram with N vertices. The endpoints carrying the coordinates $x_1, \dots, x_n$ are called exterior⁵ vertices and are denoted by $V_1, \dots, V_n$. The remaining vertices $V_{n+1}, \dots, V_N$ are called interior vertices. The lines connecting the vertices V_a, V_b will be denoted by $L_{ab\sigma}$ ($\sigma = 1, \dots, v(ab)$). V_a, V_b are called endpoints of L_{ab} connecting an exterior vertex V_a with another vertex are called exterior lines. The exterior line L_{ab} is also denoted by L_a . Lines connecting a vertex with itself will be excluded.⁶

In momentum space the unrenormalized contribution from the connected diagram $\Gamma \in \mathcal{C}_n$ has the form

$$\langle T\tilde{A}(p_1) \cdots \tilde{A}(p_n) \rangle_\Gamma^u = \frac{\delta_\Gamma}{\mathcal{S}(\Gamma)} \lim_{\epsilon \rightarrow +0} \int \frac{dk_1}{(2\pi)^4} \cdots \frac{dk_s}{(2\pi)^4} I_\Gamma$$

$$\delta_\Gamma = (2\pi)^4 \delta(p_1 + \cdots + p_n)$$
(3.3)

The symmetry number $\mathcal{S}(\Gamma)$ is

$$\mathcal{S}(\Gamma) = \sigma 2^\alpha (3!)^\beta$$
(3.4)

where α is the number of double lines (i.e., pairs of vertices connected by two lines in Fig. 1), β is the number of triple lines (pairs of vertices connected by three lines in Fig. 2), and σ is the number of automorphisms of the diagram, i.e., permutations of vertices which leave the diagram unchanged.

The integrand I_Γ is constructed according to the following insertion rules. Each line $L_{ab\sigma}$ carries the internal momentum

$$l_{ab\sigma} = -l_{ba\sigma}$$
(3.5)

To the exterior vertices $V_1, \dots, V_n$ the external momenta

$$p_1, \dots, p_n$$
(3.6)

are assigned. For an interior vertex the external momentum is set equal to zero

$$p_j = 0 \quad \text{if } j = n+1, \dots, N.$$
(3.7)

⁵ Exterior lines should be distinguished from external lines which will be introduced later in connection with subdiagrams.

⁶ Contributions from diagrams containing such loops would vanish anyhow after renormalization.

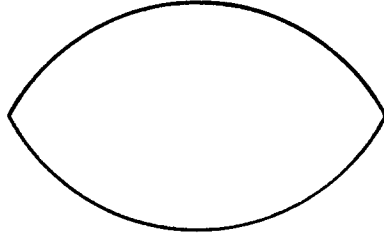


FIG. 1. Double line.

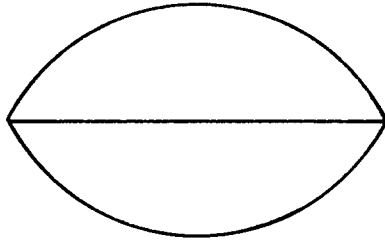


FIG. 2. Triple line.

The momenta are subject to momentum conservation at each vertex

$$\sum_{b\sigma\Gamma}^a l_{ab\sigma} = p_a, \quad a = 1, \dots, N. \quad (3.8)$$

$\sum_{b,\sigma\Gamma}^a$ denotes the sum over all lines $L_{ab\sigma}$ of Γ with V_a as one of its endpoints. The internal momenta are written as linear combinations

$$l_{ab\sigma} = l_{ab\sigma}(k) + q_{ab\sigma}(p) \quad (3.9)$$

The $q_{ab\sigma}$ are linear combinations of the external momenta $p = (p_1 \cdots p_n)$ and form a particular solution of

$$\sum_{b\sigma\Gamma}^a q_{ab\sigma} = p_a, \quad q_{ab\sigma} + q_{ba\sigma} = 0. \quad (3.10)$$

The $k_{ab\sigma}$ are linear combination of s independent internal momenta $k = (k_1 \cdots k_s)$ and form the general solution of the homogeneous equations

$$\sum_{b\sigma\Gamma}^a k_{ab\sigma} = 0, \quad k_{ab\sigma} + k_{ba\sigma} = 0 \quad (3.11)$$

With these notations the insertion rules of I_Γ are:

$$I_\Gamma = \prod_{ab\sigma\Gamma} \Delta_F^{ab\sigma} \prod_{c\Gamma} P_c \quad (3.12)$$

with

$$\Delta_F^{ab\sigma} = i(l_{ab\sigma}^2 - m^2 + i\epsilon(l_{ab\sigma}^2 + m^2))^{-1} \quad (3.13)$$

$P_c = -i\lambda$ for a 4-vertex,

$P_c = -ig$ for a 3-vertex,

$P_c = ia + ib(l^2 - m^2)$ for a 2-vertex,

($\pm l$ is the momentum of the lines joining at the 2-vertex)

$P_c = 1$ for an exterior vertex,

and $l_{ab\sigma}$ given by (3.9).

The products $\prod_{ab\sigma\Gamma}$ and $\prod_{c\Gamma}$ extend over all lines or vertices of the diagram Γ .

The renormalized contribution of the connected diagram Γ is given by the finite part of (3.3)

$$\langle T\tilde{A}(p_1) \cdots \tilde{A}(p_n) \rangle_\Gamma = \frac{\delta_\Gamma}{\mathcal{S}(\Gamma)} \lim_{\epsilon \rightarrow 0} \int \frac{dk_1}{(2\pi)^4} \cdots \frac{dk_r}{(2\pi)^4} R_\Gamma \quad (3.14)$$

R_Γ is constructed by applying Bogoliubov's subtraction rules to the original integrand I_Γ (for details see [5]). The number of subtractions used in the definition of R_Γ is specified by the degree

$$\delta(\gamma) = 4 - r(\gamma) \quad (3.15)$$

which is assigned to every proper subdiagram γ of Γ . $r(\gamma)$ denotes the number of external lines of γ (for a precise definition see the Appendix). As is shown in the Appendix the degree (3.15) satisfies conditions which are sufficient for the convergence of (3.14). In particular $\delta(\gamma)$ is greater or equal than the dimension $d(\gamma)$ of the unrenormalized integral

$$\delta(\gamma) \geq d(\gamma) = 4 - r(\gamma) - v_3(\gamma) \quad (3.16)$$

with $v_3(\gamma)$ denoting the number of 3-vertices in γ .

If the diagram Γ is not connected the contribution factorizes in the usual way. Let Γ consist of the connected components $\Gamma_1, \dots, \Gamma_c$. Some of the Γ_j may consist

of a single disconnected line. Let $p_{j1}, \dots, p_{jn_j}$ be the external momenta attached to Γ_j . Then the contribution from Γ is

$$\begin{aligned} \langle T\tilde{A}(p_1) \cdots \tilde{A}(p_n) \rangle_\Gamma &= \prod_{j=1}^c \langle T\tilde{A}(p_{j1}) \cdots \tilde{A}(p_{jn_j}) \rangle_{\Gamma_j} \\ &= \frac{\delta_\Gamma}{\mathcal{S}(\Gamma)} \lim_{\epsilon \rightarrow +0} \int \frac{dk_1}{(2\pi)^4} \cdots \frac{dk_s}{(2\pi)^4} R_\Gamma(p_1 \cdots p_n, k_1 \cdots k_s) \end{aligned} \quad (3.17)$$

where

$$\begin{aligned} \delta_\Gamma &= \prod_{j=1}^c (2\pi)^4 \delta \left(\sum_{\alpha=1}^{n_j} p_{j\alpha} \right) \\ \mathcal{S}(\Gamma) &= \prod_j \mathcal{S}(\Gamma_j), \quad R_\Gamma = \prod_j R_{\Gamma_j}, \end{aligned} \quad (3.18)$$

$k_1, \dots, k_s$ are the independent internal momenta of all nontrivial connected components.

4. DEFINITION OF COMPOSITE FIELD OPERATORS

In this section we will define composite field operators through their time-ordered functions. We consider the monomial

$$M_{\{\mu\}}\{A(x)\} = A_{(\mu)_1}(x) \cdots A_{(\mu)_m}(x) \quad (4.1)$$

with the notation

$$\begin{aligned} A_{(\mu)_j} &= \partial_{(\mu)_j} A, \\ (\mu)_j &= (\mu_{j1}, \dots, \mu_{jm(j)}), \\ \partial_{(\mu)_j} &= \partial_{\mu_{j1}} \cdots \partial_{\mu_{jm(j)}} \quad \partial_\mu = \partial/\partial x^\mu \\ \{\mu\} &= ((\mu)_1, \dots, (\mu)_m). \end{aligned} \quad (4.2)$$

The dimension of the monomial (4.1) is

$$d = d(M_{\{\mu\}}) = m + \sum_{j=1}^m \#(\mu)_j \quad (4.3)$$

$\#s$ denotes the number of elements of a set s ,

$$\#(\mu)_j = m(j).$$

We begin with the definition of the normal product

$$B_{\{u\}}(x) = N[M_{\{u\}}\{A(x)\}] \quad (4.4)$$

of minimal degree. The time ordered Green's functions of (4.4) are defined by the finite part of the expansion

$$\begin{aligned} & \langle TB_{\{u\}}(x) A(y_1) \cdots A(y_m) \rangle \\ &= \text{Fin}(\langle \text{Te}^{i[\mathcal{L}^0]_{\text{int}} dz} M_{\{u\}}\{A^0(x)\} A^0(y_1) \cdots A^0(y_m) \rangle_0 / \langle \text{Te}^{i[\mathcal{L}^0]_{\text{int}} dz} \rangle_0) \end{aligned} \quad (4.5)$$

The diagrams contributing to (4.5) contain a new vertex with the coordinate x which requires additional subtractions. The number of subtractions will be specified later. We first give the rules for constructing the unrenormalized integral of the contribution

$$\langle T\tilde{B}_{\{u\}}(p) \tilde{A}(q_1) \cdots \tilde{A}(q_n) \rangle^A \quad (4.6)$$

from a given diagram Δ . $\mathcal{C}_{m+n}$ is the set of all diagrams contributing to the Green's function

$$\langle TA(x_1) \cdots A(x_m) A(y_1) \cdots A(y_n) \rangle.$$

Let Γ be such a diagram with coordinates $x_1, \dots, x_m; y_1, \dots, y_n$ and corresponding exterior vertices $V_1, \dots, V_m; V_{m+1}, \dots, V_{m+n}$. The interior vertices are $V_{m+n+1}, \dots, V_N$. Identifying the coordinates x_j by setting $x_j = x$ we obtain a diagram contributing to (4.6). For diagrams Γ, Δ related in this way we use the notation

$$\Delta = \tilde{\Gamma}, \quad \Gamma = \tilde{\Delta}. \quad (4.7)$$

The new exterior vertex of Δ obtained by identifying $V_1, \dots, V_m$ will be denoted by V_0 . The exterior vertices of Δ are then $V_0, V_{m+1}, \dots, V_{m+n}$, the interior vertices are $V_{m+n+1}, \dots, V_N$. The set of all diagrams contributing to (4.6) will be denoted by $\mathcal{D}_{mn}$. (4.7) represents a one-to-one correspondence between $\mathcal{C}_{m+n}$ and $\mathcal{D}_{mn}$.

Let $\Gamma \in \mathcal{C}_{m+n}$ have the following connected components:

$$\Gamma_1, \dots, \Gamma_c, \quad \Gamma'_1, \dots, \Gamma'_{c'}, \quad \Gamma''_1, \dots, \Gamma''_{c''}. \quad (4.8)$$

To Γ_j at least one x_i and one y_j are attached. To Γ'_j only y 's are attached and to Γ''_j only x 's. With this the unrenormalized contribution from Γ becomes

$$\langle T\tilde{A}(p_1) \cdots \tilde{A}(p_m) \tilde{A}(q_1) \cdots \tilde{A}(q_n) \rangle^u_\Gamma = \frac{\delta_\Gamma}{\mathcal{I}(\Gamma)} \lim_{\epsilon \rightarrow +0} \int \frac{dk_1}{(2\pi)^4} \cdots \frac{dk_s}{(2\pi)^4} I_\Gamma \quad (4.9)$$

$$\delta_\Gamma = \prod_{j=1}^c (2\pi)^4 \delta \left(\sum_{\alpha=1}^{m_j} p_{j\alpha} + \sum_{\alpha=1}^{n_j} q_{j\alpha} \right) \prod_{j=1}^{c'} (2\pi)^4 \delta \left(\sum_{\alpha=1}^{n'_j} q'_{j\alpha} \right) \prod_{j=1}^{c''} (2\pi)^4 \delta \left(\sum_{\alpha=1}^{m''_j} p''_{j\alpha} \right) \quad (4.10)$$

$p_{j1}, \dots, p_{jm(j)}$ and $q_{j1}, \dots, q_{jn(j)}$ denote the external momenta of $p_1, \dots, p_m$ or $q_1, \dots, q_n$ which belong to exterior lines of the connected component Γ_j . Likewise $q'_{j1}, \dots, q'_{jn'(j)}$ denote the momenta of $q_1, \dots, q_n$ belonging to Γ'_j and $p''_{j1}, \dots, p''_{jm''(j)}$ denote the external momenta of $p_1, \dots, p_m$ which belong to Γ''_j . To simplify the notation we set

$$q_1 = p_{m+1}, \dots, q_n = p_{m+n}. \quad (4.11)$$

I_Γ is given by (3.12–13) with

$$p = (p_1, \dots, p_{m+n}). \quad (4.12)$$

A Fourier transformation with respect to p_j yields

$$\begin{aligned} & \langle TA_{(\mu)_1}(x_1) \cdots A_{(\mu)_m}(x_m) \tilde{A}(q_1) \cdots \tilde{A}(q_n) \rangle_\Gamma^u \\ &= \int \frac{dp_1}{(2\pi)^4} \cdots \frac{dp_m}{(2\pi)^4} e^{-i\sum p_j x_j} P_{\{\mu\}} \delta_\Gamma \lim_{\epsilon \rightarrow +0} \int \frac{dk_1}{(2\pi)^4} \cdots \frac{dk_s}{(2\pi)^4} I_\Gamma \end{aligned} \quad (4.13)$$

with

$$\begin{aligned} P_{\{\mu\}} &= (-i)^{\sum \#(\mu)_j} p_{1(\mu)_1} \cdots p_{m(\mu)_m} \\ \{\mu\} &= ((\mu)_1, \dots, (\mu)_m). \end{aligned} \quad (4.14)$$

Introducing new variables of integration

$$\begin{aligned} p_{j\alpha} &= \frac{p^{(j)}}{m_j} + K_{j\alpha}, & p''_{j\alpha} &= K''_{j\alpha} \\ p^{(j)} &= - \sum_{\alpha=1}^{n_j} q_{j\alpha} \end{aligned} \quad (4.15)$$

and carrying out the integration over $K_{jm_j}, K''_{jm''_j}$

$$\begin{aligned} & \langle TA_{(\mu)_1}(x_1) \cdots A_{(\mu)_m}(x_m) \tilde{A}(q_1) \cdots \tilde{A}(q_n) \rangle_\Gamma^u \\ &= \frac{\delta_{\Gamma'}}{\mathcal{S}(\Gamma)} \int \prod_j \prod_{\alpha=1}^{m_j-1} \frac{dK_{j\alpha}}{(2\pi)^4} \prod_j'' \prod_{\alpha=1}^{m_j''-1} \frac{dK''_{j\alpha}}{(2\pi)^4} e^{-i\sum p_j x_j} P_{\{\mu\}} \lim_{\epsilon \rightarrow +0} \int \prod_{j=1}^s \frac{dk_j}{(2\pi)^4} I_\Gamma \end{aligned} \quad (4.16)$$

$$\delta_{\Gamma'} = \prod_{j=1}^{e'} (2\pi)^4 \delta \left(\sum_{\alpha=1}^{n_j'} q'_{j\alpha} \right) \quad (4.17)$$

In $P_{\{\mu\}}$ and I_Γ the substitutions (4.15) are made for the external momenta. The

product $\prod_j$ is restricted to values j with $m_j \geq 1$, $\prod_j''$ is restricted to values of j with $m_j'' \geq 1$. Setting $x_j = x$ and taking a Fourier transformation with respect to x we get

$$\begin{aligned} & \langle T\tilde{M}_{\{\mu\}}(p) \tilde{A}(q_1) \cdots \tilde{A}(q_m) \rangle_{\Delta}^u \\ &= \frac{\delta_{\Delta}}{\mathcal{S}(\Delta)} \lim_{\epsilon \rightarrow +0} \int \prod_j \prod_{\alpha=1}^{m_j-1} \frac{dK_{j\alpha}}{(2\pi)^4} \prod_j'' \prod_{\alpha=1}^{m_j''-1} \frac{dK_{j\alpha}''}{(2\pi)^4} \prod_{j=1}^s \frac{dk_j}{(2\pi)^4} I_{\Delta\{\mu\}} \end{aligned} \quad (4.18)$$

where

$$\begin{aligned} \delta_{\Delta} &= (2\pi)^4 \delta \left(p + \sum q_{j\alpha} \right) \prod_{j=1}^{c'} (2\pi)^4 \delta \left(\sum_{\alpha=1}^{n_j'} q_{j\alpha}' \right) \\ \mathcal{S}(\Delta) &= \mathcal{S}(\Gamma) \end{aligned} \quad (4.19)$$

and

$$I_{\Delta\{\mu\}} = SP_{\{\mu\}} I_{\Gamma}. \quad (4.20)$$

The substitution operator S prescribes that the substitutions (4.15) be made for momenta $p_1, \dots, p_m$. According to (4.20) the insertion rules for the construction of I_{Δ} are the usual ones except that the monomial $P_{\{\mu\}}$ is assigned to the vertex V_0 .

The renormalized Green's function (4.5) is then defined by

$$\langle TB_{\{\mu\}}(x) A(y_1) \cdots A(y_n) \rangle = \sum_{\Delta \in \mathcal{G}_{mn}} \langle TB_{\{\mu\}}(x) A(y_1) \cdots A(y_n) \rangle_{\Delta} \quad (4.21)$$

with the Fourier transform of

$$\langle TB_{\{\mu\}}(x) A(y_1) \cdots A(y_n) \rangle_{\Delta} \quad (4.22)$$

given by the finite part of (4.18), namely

$$\begin{aligned} & \langle T\tilde{B}_{\{\mu\}}(p) \tilde{A}(q_1) \cdots \tilde{A}(q_n) \rangle_{\Delta} \\ &= \frac{\delta_{\Delta}}{\mathcal{S}(\Delta)} \lim_{\epsilon \rightarrow +0} \int \prod_j \prod_{\alpha=1}^{m_j-1} \frac{dK_{j\alpha}}{(2\pi)^4} \prod_j'' \prod_{\alpha=1}^{m_j''-1} \frac{dK_{j\alpha}''}{(2\pi)^4} \prod_{j=1}^s \frac{dk_j}{(2\pi)^4} R_{\Delta}^{(d)}. \end{aligned} \quad (4.23)$$

We still have to specify the number of subtractions to be used in constructing $R_{\Delta}^{(d)}$ from the unrenormalized integrand I_{Δ} . This is done by prescribing the degree $\delta_d(\gamma)$ of any proper subdiagram γ of Δ . If γ does not contain V_0 we define $\delta_d(\gamma)$ by (3.15). If γ contains V_0 we define $\delta_d(\gamma)$ by

$$\delta_d(\gamma) = d - r(\gamma) \quad (4.24)$$

where d is the dimension (4.3) of the operator $B_{\{M\}}^{(a)}$ and $r(\gamma)$ denotes the number of external lines of γ (for a precise definition of $r(\gamma)$ see the Appendix). As is shown in the Appendix $\delta_a(\gamma)$ satisfies the conditions (A.35) and (A.36) which are sufficient for the convergence of (4.23).

We will also need composite operators

$$B_{\{\omega\}}^{(a)}(x) = N_a[M_{\{\omega\}}\{A(x)\}] \quad (4.25)$$

which are constructed with additional subtractions. The degree a of the operator may be any integer which is greater or equal to the dimension (4.3)

$$a \geq d, \quad a - d = 0, 1, 2, \dots \quad (4.26)$$

The Green's functions

$$\langle TB_{\{\omega\}}^{(a)}(x) A(y_1) \cdots A(y_n) \rangle \quad (4.27)$$

are defined similarly to (4.4), (4.21–23) but with a larger number of subtractions

$$\begin{aligned} \langle TB_{\{\omega\}}^{(a)}(x) A(y_1) \cdots A(y_n) \rangle &= \sum_{\Delta \in \mathcal{Q}_{mn}} \langle TB_{\{\omega\}}^{(a)}(x) A(y_1) \cdots A(y_n) \rangle_{\Delta} \\ \langle T\tilde{B}_{\{\omega\}}^{(a)}(p) \tilde{A}(q_1) \cdots \tilde{A}(q_n) \rangle &= \frac{\delta_{\Delta}}{\mathcal{S}_{\Delta}} \lim_{\epsilon \rightarrow +0} \int \prod_j \prod_{\alpha=1}^{n_j-1} \frac{dK_{j\alpha}}{(2\pi)^4} \prod_j \prod_{\alpha=1}^{n_j-1} \frac{dK_{j\alpha}''}{(2\pi)^4} \prod_{j=1}^s \frac{dk_j}{(2\pi^4)} R_{\Delta}^{(a)} \end{aligned} \quad (4.28)$$

In order to specify the number of subtractions used in constructing $R_{\Delta}^{(a)}$ from I_{Δ} we assign a degree $\delta_a(\gamma)$ to any proper subdiagram δ of Δ . If γ does not contain V_0 we set

$$\delta_a(\gamma) = \delta(\gamma) \quad (4.29)$$

If V_0 belongs to γ we set

$$\delta_a(\gamma) = a - r(\gamma) \quad (4.30)$$

According to (4.24)

$$\delta_a(\gamma) = \delta_a(\gamma) + a - d \text{ if } V_0 \text{ belongs to } \gamma. \quad (4.31)$$

Again the degree $\delta_a(\gamma)$ satisfies the conditions (A.35), (A.36) of the Appendix which are sufficient for the convergence of (4.28).

For the derivation of some normalization conditions we write down the expansion of the proper part, as defined by (2.16–17);

$$\begin{aligned} \langle TB_{\{\mu\}}^{(a)}(x) A(y_1) \cdots A(y_n) \rangle^{\text{prop}} &= \langle T: A_{\{\mu\}_1}^0(x) \cdots A_{\{\mu\}_m}^0(x): A^0(y_1) \cdots A^0(y_n) \rangle_0^{\text{prop}} \\ &+ \sum_{\Phi \in \mathcal{F}_{mn}} \frac{1}{\mathcal{S}(\Phi)} \langle TB_{\{\mu\}}^{(a)}(x) A(y_1) \cdots A(y_n) \rangle_{\Phi}^{\text{prop}} \end{aligned} \quad (4.32)$$

The first term on the right side

$$\begin{aligned} &\langle T: A_{\{\mu\}_1}^0(x) \cdots A_{\{\mu\}_m}^0(x): A^0(y_1) \cdots A^0(y_n) \rangle_0^{\text{prop}} \\ &= \delta_{mn} \partial_{\{\mu\}_1}^{x_1} \cdots \partial_{\{\mu\}_m}^{x_m} \sum_p \delta(x_1 - y_{i_1}) \cdots \delta(x_n - y_{i_n}) |_{x_j=x} \end{aligned} \quad (4.33)$$

represents the zero order contribution and comes from the trivial diagram consisting of the vertex V_0 only. A^0 denotes a free field, $\langle \rangle_0$ the free field vacuum expectation value. The sum $\sum_{\Phi \in \mathcal{F}_{mn}}$ on the right side of (4.32) extends over the set $\mathcal{F}_{mn}$ of all proper diagrams Φ which can be obtained by dropping the lines with endpoints $y_1, \dots, y_n$ from a diagram Φ^* contributing to $\langle TB_{\{\mu\}}^{(a)}(x) A(y_1) \cdots A(y_n) \rangle$.

In momentum space the contribution from a diagram Φ is

$$\begin{aligned} &\langle T\tilde{B}_{\{\mu\}}^{(a)}(p) \tilde{A}(q_1) \cdots \tilde{A}(q_n) \rangle_{\Phi}^{\text{prop}} \\ &= \frac{\delta_{\Phi}}{\mathcal{S}(\Phi)} \lim_{\epsilon \rightarrow +0} \int \prod_j \prod_{\alpha=1}^{n_j-1} \frac{dK_{j\alpha}}{(2\pi)^4} \prod_j'' \prod_{\alpha=1}^{n_j''-1} \frac{dK_{j\alpha}''}{(2\pi)^4} \prod_{j=1}^s \frac{dk_j}{(2\pi)^4} R_{\Phi}^{(a)} \quad (4.34) \\ &\delta_{\Phi} = (2\pi)^4 \delta\left(p - \sum q_j\right), \quad \mathcal{S}(\Phi) = \mathcal{S}(\Phi^*) \end{aligned}$$

in the notation of (4.28). The renormalized integrand $R_{\Phi}^{(a)}$ can be written as

$$R_{\Phi}^{(a)} = (1 - t^{\Phi}) \bar{R}_{\Phi}^{(a)} \quad (4.35)$$

where t^{Φ} denotes the Taylor operator with respect to $q_1, \dots, q_n$ up to and including the order $\delta_a(\Phi)$. For the definition of $\bar{R}_{\Phi}^{(a)}$ see [5]. Equation (4.35) implies

$$\partial_{(v)_1}^{q_1} \cdots \partial_{(v)_n}^{q_n} R_{\Phi}^{(a)} = 0 \quad (4.36)$$

or

$$\partial_{(v)_1}^{q_1} \cdots \partial_{(v)_n}^{q_n} \langle TB_{\{\mu\}}^{(a)}(0) \tilde{A}(q_1) \cdots \tilde{A}(q_n) \rangle_{\Phi}^{\text{prop}} = 0 \quad (4.37)$$

at $q_1 = \cdots = q_n = 0$ provided

$$\sum \#(v)_i \leq \delta_a(\Phi) = a - n. \quad (4.38)$$

Hence

$$\langle TB_{\{\mu\}}^{(a)}(0) \tilde{A}_{(v)_1}(0) \cdots \tilde{A}_{(v)_n}(0) \rangle^{\text{prop}} \quad (4.39)$$

$$= \langle T: A_{\{\mu\}_1}^0(0) \cdots A_{\{\mu\}_m}^0(0): \tilde{A}_{(v)_1}^0(0) \cdots \tilde{A}_{(v)_n}^0(0) \rangle_0^{\text{prop}} \text{ provided } n + \sum \#(v)_j \leq a.$$

$$\begin{aligned} & \langle TB_{\{\mu\}}^{(a)}(0) \tilde{A}_{(v)_1}(0) \cdots \tilde{A}_{(v)_n}(0) \rangle^{\text{prop}} = 0 \\ & \text{if } n + \sum \#(v)_j \leq a \text{ and } \{v\} \not\sim \{\mu\} \end{aligned} \quad (4.40)$$

$$\begin{aligned} & \langle TB_{\{\mu\}}^{(a)}(0) \tilde{A}_{(v)_1}(0) \cdots \tilde{A}_{(v)_n}(0) \rangle^{\text{prop}} = i^{\sum \#(\mu)_j} \prod_{j=1}^t s_j! \prod_{j=1}^m \prod_{\alpha=0}^3 a_{j\alpha}! \\ & \text{if } n + \sum \#(v)_j \leq a \text{ and } \{v\} \sim \{\mu\}. \end{aligned}$$

In these equations the following notations were used. Two sets $(\mu)_i, (v)_j$ are called equivalent

$$(\mu)_i \sim (v)_j \quad (4.41)$$

if $(v)_j$ is a permutation of $(\mu)_i$. $\{\mu\}$ and $\{v\}$ are called equivalent

$$\{\mu\} \sim \{v\} \quad (4.42)$$

if $n = m$ and if there is a permutation $((v)_{i_1}, \dots, (v)_{i_n})$ of $\{v\}$ such that

$$(\mu)_j \sim (v)_{i_j}$$

Otherwise we write $\{\mu\} \not\sim \{v\}$, $s_1, \dots, s_t$ are the numbers of equivalent $(\mu)_k$ occurring in $\{\mu\}$. $a_{j\alpha}$ is the number of α 's occurring in $(\mu)_j$.

We finally show that the vacuum expectation values

$$\langle B_{\{\mu\}}^{(a)}(x) \rangle = 0 \quad (4.43)$$

vanish. The contribution from a Feynman diagram Δ is given by

$$\langle \tilde{B}_{\{\mu\}}^{(a)}(p) \rangle_{\Delta} = \frac{\delta(p)}{\mathcal{P}(\Delta)} \lim_{\epsilon \rightarrow +0} \int \prod_j \prod_{\alpha=1}^{n_j-1} \frac{dK_{j\alpha}''}{(2\pi)^4} \prod_{j=1}^s \frac{dk_j}{(2\pi)^4} R_{\Delta}^{(a)} \quad (4.44)$$

If Δ is proper the integrand is of the form

$$R_{\Delta}^{(a)} = (1 - t_p^a) \bar{R}_{\Delta}^{(a)} = 0$$

and vanishes since $R^{(a)}$ does not depend on p . If Δ is not proper it contains at least

one proper subdiagram γ with one external momentum p only and degree $b = \delta(\gamma) \geq 0$. Such a diagram contributes to $R_\Delta^{(a)}$ the factor

$$R_\gamma^{(b)} = (1 - t_{p\gamma}^b) \bar{R}_\gamma^{(b)} = 0$$

since $\bar{R}_\gamma^{(b)}$ does not depend on p^γ .

As has been emphasized by Lowenstein [13] the time ordered Green's functions defined in this section are not uniquely determined. It frequently occurs that the same local operator $B(x)$ may be represented by different polynomials of $A(x)$

$$B(x) = \sum c_j N_{a_j} [M_j \{A(x)\}] \quad (4.45)$$

$$B(x) = \sum c'_j N_{a'_j} [M'_j \{A(x)\}] \quad (4.46)$$

M_j , M'_j denote monomials of A and its derivatives, j includes Lorentz indices. While the time ordered functions of a given monomial were uniquely prescribed the Green's function

$$\langle TB(x) A(y_1) \cdots A(y_n) \rangle \quad (4.47)$$

as calculated from the representation (4.45) and (4.46) may differ by δ -function terms.

5. ALGEBRAIC IDENTITIES

Let Γ be a diagram of $\mathcal{C}_{m+n}$ and $\Delta = \tilde{\Gamma} \in \mathcal{D}_{mn}$. In this section we will investigate the relation between R_Γ and the integrand $R_\Delta^{(a)}$ which we used in (4.28) for defining the Green's functions of composite operators. Using the explicit formulas derived in [5, Section 3], R_Γ and $R_\Delta^{(a)}$ may be written as

$$R_\Gamma = S_\Gamma \sum_{U \in \mathcal{W}(\Gamma)} \prod_{\gamma \in U} (-t^\gamma S_\gamma) S_U I_\Gamma \quad (5.1)$$

$$R_\Delta^{(a)} = S_\Delta \sum_{U \in \mathcal{W}_a(\Gamma)} \prod_{\gamma \in U} (-t_a^\gamma S_\gamma) S_U I_\Delta \quad (5.2)$$

I_Γ and I_Δ are related by (4.20). $U(\Gamma)$ is the set of all Γ -forests, $U_a(\Delta)$ the set of all Δ -forests relative to the degree $\delta_a(\gamma)$. For the definition of the substitution operators S_γ see [5]. In addition we introduce the substitution operator S_U . If applied to a function of the momenta $l_{ab\sigma}(p, k)$ the operator S_U prescribes the substitutions

$$l_{ab\sigma}(pk) \rightarrow l_{ab\sigma}'(p^\gamma k^\gamma)$$

if $L_{ab\sigma}$ belongs to $\gamma \in U$ but not to any other $\gamma' \in U$ with $\gamma' \subset \gamma$.

Apparently

$$\mathcal{U}_a(\Delta) = \mathcal{U}(\Gamma) \cup \Phi \quad (5.3)$$

where Φ is the set of all Δ -forests with at least one element γ containing V_0 . From (5.1–3)

$$R_\Delta^{(a)} = SP_{(u)} R_\Gamma + X \quad (5.4)$$

with

$$X = S_\Delta \sum_{U \in \Phi} \prod_{\gamma \in U} (-t_a^\gamma S_\gamma) S_U I_\Delta \quad (5.5)$$

Our aim is to write X in a more convenient form. First we note that any two elements γ, γ' of U containing V_0 must satisfy

$$\gamma \subset \gamma' \text{ or } \gamma' \subset \gamma$$

For γ and γ' are not disjoint since they have the vertex V_0 in common. Hence among all elements $\gamma \in U$ containing V_0 there is a smallest one which we call τ . Then

$$U = U_1 \cup U_2 \cup \{\tau\} \quad (5.6)$$

U_1 is the set of all $\gamma \in U (\gamma \neq \tau)$ with $\tau \subset \gamma$ or τ and γ disjoint. U_2 is the set of all $\gamma \in U (\gamma \neq \tau)$ with $\gamma \subset \tau$. (5.6) implies

$$\prod_{\gamma \in U} (-t_a^\gamma S_\gamma) = \prod_{\gamma \in U_1} (-t_a^\gamma S_\gamma) (-t_a^\tau S_\tau) \prod_{\gamma \in U_2} (-t^\gamma S_\gamma) \quad (5.7)$$

With this information Eq. (5.5) becomes

$$X = \sum_{\tau \in T_a} \sum_{U_1 \in \mathcal{M}_\tau} \sum_{U_2 \in \mathcal{U}(\tau)} \prod_{\gamma \in U_1} (-t_a^\gamma S_\gamma) (-t_a^\tau S_\tau) \prod_{\gamma \in U_2} (-t^\gamma S_\gamma) S_U I_\Delta \quad (5.8)$$

Here T_a denotes the set of all proper subdiagrams τ of Δ which contain V_0 and have degree $\delta_a(\tau) \geq 0$. $\mathcal{M}_\tau$ is the set of all Δ -forests U_1 which have the property that each $\gamma \in U_1$ satisfies

$$\tau \subset \gamma \quad \text{or} \quad \tau \cap \gamma = \emptyset.$$

$\mathcal{U}(\tau)$ is the set of all τ -forests. (5.4) and (5.8) represent the final form of the algebraic identities.

We finally discuss a generalization of (5.4), (5.8) which will be needed in the following section for establishing relations between normal products of different

degrees. Suppose $R_\Delta^{(a)}$ and $R_\Delta^{(b)}$ are formed with different degrees δ_a and δ_b defined through (4.29–30), assuming $a > b$. We have

$$R_\Delta^{(a)} = S_\Delta \sum_{U \in \mathcal{U}_\Delta(\Delta)} \prod_{\gamma \in U} (-t_a^\gamma S_\gamma) S_U I_\Delta \quad (5.9)$$

$$R_\Delta^{(b)} = S_\Delta \sum_{U \in \mathcal{U}_\Delta(\Delta)} \prod_{\gamma \in U} (-t_b^\gamma S_\gamma) S_U I_\Delta \quad (5.10)$$

In (5.10) we used the convention

$$t_b^\gamma = 0 \quad \text{if } \delta_b(\gamma) < 0.$$

For any γ containing V_0 we split

$$t_a^\gamma = t_b^\gamma + (t_a^\gamma - t_b^\gamma) \quad (5.11)$$

while

$$t_a^\gamma = t_b^\gamma = t_{p^\gamma}^{d(\gamma)} \quad (5.12)$$

holds for all γ 's which do not contain V_0 . Substituting (5.11) into (5.9) we obtain

$$R_\Delta^{(a)} = S_\Delta \sum_{U \in \mathcal{U}_\Delta(\Delta)} \sum_{F \in \mathcal{F}(U)} \prod_{\gamma \in U} F_\gamma S_U I_\Delta \quad (5.13)$$

Here $\mathcal{F}(U)$ is the family of functions with the property

- (i) $F_\gamma = -t_b^\gamma S_\gamma$ or $F_\gamma = -(t_a^\gamma - t_b^\gamma) S_\gamma$
if γ contains V_0 ,
- (ii) $F_\gamma = -t_a^\gamma S_\gamma = -t_b^\gamma S_\gamma$
if γ does not contain V_0 .

For any U there is a function F_0 in $\mathcal{F}(U)$ which assigns $F_\gamma \equiv -t_b^\gamma S_\gamma$ to any $\gamma \in U$. Taking out all terms with $F = F_0$ we find

$$R_\Delta^{(a)} = R_\Delta^{(b)} + X \quad (5.14)$$

$$X = S_\Delta \sum_{U \in \mathcal{U}'} \sum_{\substack{F \in \mathcal{F}(U) \\ F \neq F_0}} \prod_{\gamma \in U} F_\gamma S_U I_\Delta \quad (5.15)$$

$\mathcal{U}'$ is the set of all forests having at least one element γ which contains V_0 . Our aim is again to write X in a convenient form. For $F \neq F_0$ there is a smallest element τ in U which contains V_0 and satisfies

$$F_\tau = -(t_a^\tau - t_b^\tau) S_\tau \quad (5.16)$$

For given U and F we decompose

$$U = U_1 \cup U_2 \cup \{\tau\} \quad (5.17)$$

where U_1 and U_2 are defined as below Eq. (5.6). As generalization of (5.4), (5.8) we finally obtain

$$R_\Delta^{(a)} = R_\Delta^{(b)} + X \quad (5.18)$$

$$X = \sum_{\tau \in T_\Delta} \sum_{U_1 \in \mathcal{M}_\tau} \sum_{U_2 \in \mathcal{U}(\tau)} \prod_{\gamma \in U_1} (-t_a^\gamma S_\gamma) (-t_a^\tau - t_b^\tau) S_\tau \prod_{\gamma \in U_2} (-t_b^\gamma S_\gamma) S_U I_\Delta$$

$T_a, \mathcal{M}_\tau$ are defined as below Eq. (5.8).

6. RELATIONS BETWEEN COMPOSITE OPERATORS OF DIFFERENT DEGREE

In this section we will derive some identities which relate composite operators of different degree. We consider the formal operator product

$$M_{\{\mu\}}\{A(x)\} = A_{\{\mu\}_1}(x) \cdots A_{\{\mu\}_m}(x), \quad m \geq 2 \quad (6.1)$$

of dimension

$$d = m + \sum \#(\mu)_j \geq 2 \quad (6.2)$$

and form the composite operators

$$B_{\{\mu\}}^{(a)}(x) = N_a[M_{\{\mu\}}\{A(x)\}] \quad (6.3)$$

$$B_{\{\mu\}}^{(b)}(x) = N_b[M_{\{\mu\}}\{A(x)\}]. \quad (6.4)$$

It will always be assumed that

$$d \leq b < a. \quad (6.5)$$

We will compare the power series expansions of the Green's functions

$$\langle TB_{\{\mu\}}^{(a)}(x) A(y_1) \cdots A(y_n) \rangle = \sum_{\Delta \in \mathcal{Q}_{mn}} \langle TB_{\{\mu\}}^{(a)}(x) A(y_1) \cdots A(y_n) \rangle_\Delta \quad (6.6)$$

$$\langle TB_{\{\mu\}}^{(b)}(x) A(y_1) \cdots A(y_n) \rangle = \sum_{\Delta \in \mathcal{Q}_{mn}} \langle TB_{\{\mu\}}^{(b)}(x) A(y_1) \cdots A(y_n) \rangle_\Delta. \quad (6.7)$$

According to (4.28) the contributions from the diagram Δ are given by

$$\begin{aligned} & \langle T\tilde{B}_{\{\mu\}}^{(a)}(p) \tilde{A}(q_1) \cdots \tilde{A}(q_n) \rangle_{\Delta} \\ &= \frac{\delta_{\Delta}}{\mathcal{S}(\Delta)} \lim_{\epsilon \rightarrow +0} \int \prod_j \prod_{\alpha=1}^{m_j-1} \frac{dK_{j\alpha}}{(2\pi)^4} \prod_j'' \prod_{\alpha=1}^{n_j''-1} \frac{dK_{j\alpha}''}{(2\pi)^4} \prod_{j=1}^s \frac{dk_j}{(2\pi)^4} R_{\Delta}^{(a)} \end{aligned} \quad (6.8)$$

$$\begin{aligned} & \langle T\tilde{B}_{\{\mu\}}^{(b)}(p) \tilde{A}(q_1) \cdots \tilde{A}(q_n) \rangle_{\Delta} \\ &= \frac{\delta_{\Delta}}{\mathcal{S}(\Delta)} \lim_{\epsilon \rightarrow +0} \int \prod_j \prod_{\alpha=1}^{m_j-1} \frac{dK_{j\alpha}}{(2\pi)^4} \prod_j'' \prod_{\alpha=1}^{n_j''-1} \frac{dK_{j\alpha}''}{(2\pi)^4} \prod_{j=1}^s \frac{dk_j}{(2\pi)^4} R_{\Delta}^{(b)}. \end{aligned} \quad (6.9)$$

The renormalized integrands $R_{\Delta}^{(a)}$ and $R_{\Delta}^{(b)}$ are related by the identity (5.14–15). We first determine the set T_a appearing in (5.15). By definition T_a is the set of all proper subdiagrams of Δ which contain V_0 and have nonnegative degree

$$\delta_a(\tau) \geq 0. \quad (6.10)$$

The definition (4.24) of the degree yields the inequality

$$a - r \geq 0 \quad (6.11)$$

where r is the number of external lines of τ . This implies

$$r = 1, \dots, a \quad (6.12)$$

The case $r = 0$ may be excluded since

$$(t_a^{\tau} - t_b^{\tau}) S_{\tau} \prod_{\gamma \in U_2} (-t_b^{\gamma} S_{\gamma}) S_U I_{\Delta} = 0$$

vanishes if τ has no external lines. For then all momenta $1_{ab\sigma}^{\gamma}$ are independent of the external momenta p^{τ} . Hence T_a is the set of all proper subdiagrams of which contain V_0 and have $r = 1, \dots, a$ external lines.

The unrenormalized integral I_{Δ} factorize according to

$$I_{\Delta} = I_{\Delta/\tau} I_{\tau} \quad (\text{see Eq. (A.26)}). \quad (6.13)$$

With (6.13) the identity (5.14–15) becomes

$$\begin{aligned} R_{\Delta}^{(a)} &= R_{\Delta}^{(b)} + X \\ X &= - \sum_{\tau \in T_a} \sum_{U_1 \in \mathcal{A}_{\tau}} \sum_{U_2 \in \mathcal{U}(\tau)} \prod_{\gamma \in U_1} (-t_a^{\gamma} S_{\gamma}) \\ &\quad \times S_{U_1} I_{\Delta/\tau} (t_a^{\tau} - t_b^{\tau}) S_{\tau} \prod_{\gamma \in U_2} (-t_b^{\gamma} S_{\gamma}) S_{U_2} I_{\tau}. \end{aligned} \quad (6.14)$$

In order to work out the explicit form of the differential operator $t_a^\tau - t_b^\tau$ we list all nonvanishing external momenta of τ . The external lines of τ are given by (A.2). Some of the subscripts a_j may be zero. In this case external lines of τ are attached to V_0 . The nonvanishing external momenta of τ are then

$$p_0^\tau = p_0 = p$$

and

$$p_1^\tau, \dots, p_r^\tau$$

corresponding to the external lines (A.2). The momenta $p_{m+1} = q_1, \dots, p_{m+n} = q_n$ cannot be external to τ . For the corresponding exterior vertices $V_{m+1}, \dots, V_{m+n}$ do not belong to τ since τ is proper. The sum of all external momenta of τ must vanish

$$p_0^\tau + \sum_{j=1}^r p_j^\tau = 0 \quad (6.15)$$

Since τ is connected (6.15) is the only linear relation among the external momenta. Choosing $p_1^\tau, \dots, p_r^\tau$ as linearly independent external momenta we have

$$t_a^\tau - t_b^\tau = \sum_{\{\rho\}}^a \frac{1}{\prod \#(\rho)_j!} p_{1(\rho)_1}^\tau \cdots p_{r(\rho)_r}^\tau \theta^\tau \partial_{p_1^\tau}^{(\rho)_1} \cdots \partial_{p_r^\tau}^{(\rho)_r} \quad (6.16)$$

with

$$\begin{aligned} (\rho)_j &= (\rho_{j1}, \dots, \rho_{jr(j)}), & \{\rho\} &= ((\rho)_1, \dots, (\rho)_r) \\ p_{j(\rho)_j}^\tau &= p_{j\rho_1}^\tau \cdots p_{j\rho_{jr(j)}}^\tau, & \partial_{p_j^\tau}^{(\rho)_j} &= \partial_{p_{j1}^\tau}^{\rho_{j1}} \cdots \partial_{p_{jr(j)}^\tau}^{\rho_{jr(j)}} \\ \partial_{p_k^\tau}^{\rho_{jk}} &= \frac{\partial}{\partial (p_k^\tau)_{\rho_{jk}}}. \end{aligned} \quad (6.17)$$

The sum $\sum_{\{\rho\}}^a$ extends over all sets $\{\rho\}$ with

$$b < r + \sum_{j=1}^r \#(\rho)_j \leq a \quad (6.18)$$

The operator θ^τ prescribes that the external momenta p_j^τ be set equal to zero. Let S_μ be the first substitution operator of

$$S_\Gamma \prod_{\gamma \in U_1} (-t_a^\gamma S_\gamma)$$

to act on (6.16). That is, μ is either Γ or the smallest element of U_1 which contains τ . S_μ then effects the substitution

$$p_j^\tau \rightarrow p_j^\tau(k^\mu, p^\mu)$$

with

$$p_j^\tau(k^\mu, p^\mu) = -l_{a_j b_j \sigma_j}^\mu = -k_{a_j b_j \sigma_j}^\mu(k^\mu) - q_{a_j b_j \sigma_j}^\mu(p^\mu) \quad (6.19)$$

In (6.14) we may therefore replace

$$\cdots S_\mu \cdots p_{1(\rho)_1}^\tau \cdots p_{r(\rho)_r}^\tau \cdots$$

by

$$\cdots S_\mu \cdots (-il_{a_1 b_1 \sigma_1}^\mu)_{(\rho)_1} \cdots (-il_{a_r b_r \sigma_r}^\mu)_{(\rho)_r} (-i)^r \cdots.$$

We thus obtain the following factorization of X :

$$X = - \sum_{\tau \in T_a} \sum_{\{b\}}^a \frac{(-i)^{\sum \#(\rho)_j}}{\prod \#(\rho)_j!} R_{\Delta/\tau(\rho)}^{(a)} \{ \partial_{p_1}^{(\rho)_1} \cdots \partial_{p_m}^{(\rho)_m} R_\tau^{(b)} \}_{p_j^{\tau=0}} \quad (6.20)$$

$$\begin{aligned} R_{\Delta/\tau(\rho)}^{(a)} &= \sum_{U_1 \in \mathcal{H}_\tau} \prod_{\gamma \in U_1} (-t_a^\gamma S_\gamma) \prod_{j=1}^r l_{a_j b_j \sigma_j}^\mu(k^\mu, p^\mu)_{(\rho)_j} S_{U_1} I_{\Delta/\tau(\rho)} \\ &= \sum_{V \in \mathcal{F}(\Delta/\tau)} \prod_{\rho \in V} (-t_a^\rho S_\rho) S_V I_{\Delta/\tau(\rho)} \end{aligned} \quad (6.21)$$

$$R_\tau^{(b)} = \sum_{U \in \mathcal{F}(\tau)} \prod_{\gamma \in U} (-t_b^\gamma S_\gamma) S_U I_\tau. \quad (6.22)$$

In (6.21) the substitution $\rho = \gamma/\tau$ was used. For the notion of the reduced diagram Δ/τ and related definitions see the Appendix. $I_{\Delta/\tau(\rho)}$ is defined by

$$I_{\Delta/\tau(\rho)} = \prod_{j=1}^r l_{0b_j \sigma_j}^{\Delta/\tau}(k, p)_{(\rho)_j} I_{\Delta/\tau} \quad (6.23)$$

Integrating (6.20) over the internal momenta, summing over all Δ and taking a Fourier transformation with respect to $p, q_1, \dots, q_n$ we obtain

$$\begin{aligned} \langle TB_{\{\mu\}}^{(b)}(x) A(y_1) \cdots A(y_n) \rangle &= \langle TB_{\{\mu\}}^{(a)}(x) A(y_1) \cdots A(y_n) \rangle \\ &+ \sum_{\{b\}}^a \frac{(-i)^{\sum \#(\rho)_j}}{\prod \#(\rho)_j!} G_{\{\mu\}}^{(b)\{\rho\}} \langle TB_{\{\rho\}}^{(a)}(x) A(y_1) \cdots A(y_n) \rangle \end{aligned} \quad (6.24)$$

$$b < r + \sum_{j=1}^r \#(\rho)_j \leq a.$$

The coefficients are

$$G_{\{\mu\}}^{(b)\{\rho\}} = \sum_{\tau \in \mathcal{F}_r} \langle TB_{\{\mu\}}^{(b)}(0) \tilde{A}^{(\rho)_1}(0) \cdots \tilde{A}^{(\rho)_r}(0) \rangle_\tau^{\text{prop}} \quad (6.25)$$

where $\mathcal{F}_r$ is the set of all nontrivial diagrams contributing to the proper part. The notation in (6.25) is similar to (2.24), i.e.,

$$\begin{aligned} & \langle TB_{\{\mu\}}^{(b)}(x) \tilde{A}^{(\rho)_1}(q_1) \cdots \tilde{A}^{(\rho)_r}(q_r) \rangle^{\text{prop}} \\ &= \partial_{q_1}^{(\rho)_1} \cdots \partial_{q_r}^{(\rho)_r} \langle TB_{\{\mu\}}^{(b)}(x) \tilde{A}(q_1) \cdots \tilde{A}(q_r) \rangle^{\text{prop}} \end{aligned} \quad (6.26)$$

The contribution

$$\langle T: A_{0(\mu)_1}(0) \cdots A_{0(\mu)_m}(0): \tilde{A}_0^{(\rho)_1}(0) \cdots \tilde{A}_0^{(\rho)_r}(0) \rangle^{\text{prop}}$$

from the trivial diagram (with the notation of (2.24)) vanishes because

$$d = m + \sum \#(\mu)_j \leq b < r + \sum \#(\rho)_j$$

in (6.24), and therefore $\{\rho\} \not\sim \{\mu\}$. Hence

$$G_{\{\mu\}}^{(b)\{\rho\}} = \langle TB_{\{\mu\}}^{(b)}(0) \tilde{A}^{(\rho)_1}(0) \cdots \tilde{A}^{(\rho)_r}(0) \rangle^{\text{prop}} \quad (6.27)$$

for the coefficients of (6.24) with the notation (6.26). According to this the $G_{\mu}^{(b)\{\rho\}}$ are invariant tensors which can be composed of $g_{\mu\nu}$ - and $\epsilon_{\mu\nu\rho\sigma}$ -tensors only.

Applying the reduction technique to the variables $y_1, \dots, y_n$ of Eq. (6.24) we obtain the operator identity

$$\begin{aligned} B_{\{\mu\}}^{(b)}(x) &= B_{\{\mu\}}^{(a)}(x) + \sum_{\{\rho\}}^a \frac{(-i)^{\sum \#(\rho)_j}}{r! \prod \#(\rho)_j!} G_{\{\mu\}}^{(b)\{\rho\}} B_{\{\rho\}}^{(a)}(x) \\ b &< r + \sum_{j=1}^r \#(\rho)_j \leq a \end{aligned} \quad (6.28)$$

with the coefficients given by (6.27). If the reduction technique is applied to only part of the variables y_j we obtain (6.24) in operator form

$$\begin{aligned} & TB_{\{\mu\}}^{(b)}(x) A(y_1) \cdots A(y_n) \\ &= TB_{\{\mu\}}^{(a)}(x) A(y_1) \cdots A(y_n) + \sum_{\{\rho\}}^a \frac{(-i)^{\sum \#(\rho)_j}}{r! \prod \#(\rho)_j!} G_{\{\mu\}}^{(b)\{\rho\}} TB_{\{\rho\}}^{(a)}(x) A(y_1) \cdots A(y_n) \\ b &< r + \sum_{j=1}^r \#(\rho)_j \leq a. \end{aligned} \quad (6.29)$$

The coefficients $G_{\{\mu\}}^{(b)\{\rho\}}$ can also be calculated directly from (6.24) and the normalization conditions (4.40) of the normal products.

We finally derive a formula which expresses $B_{\{\mu\}}^{(b)}$ in terms of the operators $B_{\{\rho\}}$ of minimal degree. To this end we write (6.28) for $a = b + 1$

$$B_{\{\mu\}}^{(b)} = B_{\{\mu\}}^{(b+1)} + \sum_{\{\rho\}}^{b+1} \frac{(-i)^{\sum \#(\rho)_j}}{r! \prod \#(\rho)_j!} G_{\{\mu\}}^{(b)\{\rho\}} B_{\{\rho\}}^{(b+1)}. \quad (6.30)$$

The sum is restricted by

$$b < r + \sum_{j=1}^r \#(\rho)_j \leq b + 1.$$

Hence the degree $b + 1$ equals the dimension of the composite operators appearing in the sum

$$r + \sum_{j=1}^r \#(\rho)_j = b + 1$$

$$B_{\{\rho\}}^{(b+1)} = B_{\{\rho\}}.$$

With the abbreviation

$$G_{\{\mu\}}^{(b)\{\rho\}} = G_{\{\mu\}}^{\{\rho\}} \quad \text{if} \quad b = r + \sum_{j=1}^r \#(\rho)_j - 1 \quad (6.31)$$

we find

$$B_{\{\mu\}}^{(a)} = B_{\{\mu\}} - \sum_{\{\rho\}}^a \frac{(-i)^{\sum \#(\rho)_j}}{r! \prod \#(\rho)_j!} G_{\{\mu\}}^{\{\rho\}} B_{\{\rho\}}. \quad (6.32)$$

Here d denotes the dimension of $B_{\{\mu\}}$

$$d = m + \sum_{j=1}^m \#(\mu)_j.$$

The sum is restricted by

$$d < r + \sum \#(\rho)_j \leq a \quad (6.33)$$

A corresponding formula holds for time ordered products

$$\begin{aligned} \langle TB_{\{\mu\}}^{(a)}(x) A(y_1) \cdots A(y_n) \rangle &= \langle TB_{\{\mu\}}(x) A(y_1) \cdots A(y_n) \rangle \\ &\quad - \sum_{\{\rho\}}^a \frac{(-i)^{\sum \#(\rho)_j}}{r! \prod \#(\rho)_j!} G_{\{\mu\}}^{\{\rho\}} \langle TB_{\{\rho\}}(x) A(y_1) \cdots A(y_n) \rangle \end{aligned} \quad (6.34)$$

Many examples for the relations (6.24–34) can be found in the applications of the normal product algorithm by Lowenstein and Schroer [18, 24–26].

APPENDIX: SUBDIAGRAM AND REDUCED DIAGRAMS

The conventions concerning subdiagrams and reduced diagrams used in this paper are slightly different from those of [5]. We briefly introduce the relevant modifications.

A subdiagram γ of $\Gamma \in \mathcal{C}_n$ or $\Delta \in \mathcal{D}_{mn}$

$$\gamma \subset \Gamma \quad \text{or} \quad \gamma \subset \Delta$$

is defined by a nonempty subset of lines of Γ .⁷ The vertices of γ are formed by all endpoints of lines belonging to γ . A line of Γ or Δ not belonging to γ is called an external line of γ if one or both endpoints lie in γ .

We next introduce external and internal momenta for γ . Let

$$V_{c_1}, \dots, V_{c_f} \tag{A.1}$$

be the vertices and

$$L_{a_1 b_1 \sigma_1}, \dots, L_{a_r b_r \sigma_r} \tag{A.2}$$

be the external lines of γ . The indices a_j, b_j are ordered such that a_j refers to the endpoint V_{a_j} which belongs to γ . If both endpoints V_a and V_b of a line $L_{ab\sigma}$ belong to γ this line is listed twice in (A.2) as $L_{ab\sigma}$ and $L_{ba\sigma}$.

As external momenta of γ we introduce

$$p_{c_1}, \dots, p_{c_f} \tag{A.3}$$

(corresponding to each vertex (A.4) of γ) and the new variables

$$p_1^\gamma, \dots, p_r^\gamma \tag{A.4}$$

corresponding to each external line of γ as listed in (A.2). The combined sets (A.3) and (A.4) of variables will be denoted by the symbol

$$p^\gamma = (p_{c_1}, \dots, p_{c_f}, p_1^\gamma, \dots, p_r^\gamma) \tag{A.5}$$

The momenta p_{c_j} associated with an interior vertex V_{c_j} of Γ or Δ vanish. Hence only those momenta p_j in (A.3) can be different from zero which belong to the exterior vertices

$$V_1, \dots, V_n \quad \text{of} \quad \Gamma \in \mathcal{C}_n \tag{A.6}$$

⁷ The empty set or a trivial diagram consisting of a single vertex and no line is not a subdiagram in the sense of this definition. On the other hand the full diagram itself is considered as a subdiagram.

or

$$V_0, V_{m+1}, \dots, V_{m+n} \quad \text{of} \quad \Delta \in \mathcal{D}_{mn}. \quad (\text{A.7})$$

At each vertex V_a of γ we introduce the total external momentum

$$\begin{aligned} q_a^\gamma &= q_a^\gamma(p^\gamma) = p_a + \sum_{\gamma'}^{a^{\text{ext}}} p_j^\gamma \\ a &= c_1, \dots, c_{N(\gamma)} \end{aligned} \quad (\text{A.8})$$

The sum extends over all external lines $L_{ab_j\sigma_j}$ of γ which are attached to V_a . These are all lines listed in (A.2) with $a_j = a$.

The internal momenta $l_{ab\sigma}^\gamma$ are required to satisfy momentum conservation at each vertex of γ

$$\sum_{\gamma'}^a l_{ab\sigma}^\gamma = q_a^\gamma(p^\gamma) \quad (\text{A.9})$$

The $l_{ab\sigma}^\gamma$ will be written in the form

$$l_{ab\sigma}^\gamma = k_{ab\sigma}^\gamma(k^\gamma) + q_{ab\sigma}^\gamma(p^\gamma) \quad (\text{A.10})$$

where the $q_{ab\sigma}^\gamma(p^\gamma)$ are linear combinations of the external momenta p^γ and form a particular solution of

$$\sum_{\gamma'}^a q_{ab\sigma}^\gamma = q_a^\gamma(p^\gamma), \quad q_{ab\sigma}^\gamma + q_{ba\sigma}^\gamma = 0. \quad (\text{A.11})$$

The $k_{ab\sigma}^\gamma$ are linear combinations of $s(\gamma)$ independent internal momenta

$$k^\gamma = (k_1^\gamma, \dots, k_s^\gamma)_{(\gamma)}$$

and form the general solution of

$$\sum_{\gamma'}^a k_{ab\sigma}^\gamma = 0, \quad k_{ab\sigma}^\gamma + k_{ba\sigma}^\gamma = 0. \quad (\text{A.12})$$

k^γ and p^γ are expressed in terms of k and p

$$k^\gamma = k^\gamma(k, p), \quad p^\gamma = p^\gamma(k, p) \quad (\text{A.13})$$

by requiring

$$l_{ab\sigma}^\gamma(k^\gamma, p^\gamma) \equiv l_{ab\sigma}(k, p) \quad (\text{A.14})$$

for each internal line of γ and

$$p_{a_j}^\gamma = -l_{ab\sigma}(k, p) \quad (\text{A.15})$$

for each external line of γ .

Let μ be a subdiagram of $\gamma \subset \Gamma$. In an analogous way we introduce the functions

$$k^\mu = k_\nu^\mu(k^\nu, p^\nu), \quad p^\mu = p_\nu^\mu(k^\nu, p^\nu) \quad (\text{A.16})$$

by setting

$$l_{ab\sigma}^\mu(k^\mu, p^\mu) \equiv l_{ab\sigma}^\gamma(k^\gamma, p^\gamma) \quad (\text{A.17})$$

for every line $L_{ab\sigma}$ of μ . Moreover, we require

$$p_j^\mu = -l_{a_j b_j \sigma_j}^\gamma(k^\gamma, p^\gamma) \quad (\text{A.18})$$

if $L_{a_j b_j \sigma_j}$ is an external line of μ , but not of γ , and

$$p_j^\mu = p_k^\gamma \quad (\text{A.19})$$

if $L_{a_j b_j \sigma_j} = L_{a_k b_k \sigma_k}$ is an external line of μ and γ . (The numbering (A.2) of external lines may be different for μ and γ , hence $j \neq k$ in general.)

The choice of the particular solutions $q_{ab\sigma}(p)$ of (A.11) should be such that k^ν depends on k only and k^μ on k^ν only in (A.13) and (A.16)

$$k = k(k^\nu), \quad k^\mu = k_\nu^\mu(k^\nu). \quad (\text{A.20})$$

As was shown in [5] such $q_{ab\sigma}(p)$, $q_{ab\sigma}^\gamma(p^\gamma)$ can always be constructed.

The unrenormalized Feynman integral associated with a subdiagram γ of $\Gamma \in \mathcal{C}_n$ or $\Delta \in \mathcal{D}_{mn}$ is given by

$$J_\gamma = \int dk_1^\gamma \cdots dk_{s(\gamma)}^\gamma I_\gamma \quad (\text{A.21})$$

where

$$I_\gamma = \prod_{ab\sigma} \Delta_{Fab\sigma} \prod_c P_c \quad (\text{A.22})$$

with the usual insertion rules. The products extend over all lines and vertices of γ . The dimension $d(\gamma)$ is defined by the dimension of the integral (A.21).

A diagram is called trivial if it consists of a single vertex and no line. A diagram is called proper if it cannot be separated in two parts by cutting a single line. Two diagrams are called disjoint if they have neither a line nor a vertex in common. Let γ be a subdiagram of $\Gamma \in \mathcal{C}_n$ or $\Delta \in \mathcal{D}_{mn}$. Let $\gamma_1, \dots, \gamma_c$ be mutually disjoint proper subdiagrams of γ . The reduced diagram

$$\bar{\gamma} = \gamma / \gamma_1 \cdots \gamma_c \quad (\text{A.23})$$

is then defined by reducing each γ_j in γ to a vertex. In other words, the lines of $\bar{\gamma}$

are formed by those lines of γ which do not belong to any γ_j . The vertices which are formed by the corresponding endpoints but with all vertices identified which belong to the same γ_j . The unrenormalized Feynman integral associated with (A.23) is given by

$$J_{\bar{\gamma}} = \int dk_1^\gamma \cdots dk_{s(\bar{\gamma})}^\gamma I_{\bar{\gamma}} \quad (\text{A.24})$$

where

$$I_{\bar{\gamma}} = \prod_{ab\sigma} \Delta_{Fab\sigma} \prod_c P_c \quad (\text{A.25})$$

is formed with the usual insertion rules. The products extend over the lines and vertices of γ which do not belong to any γ_j . Accordingly the factor 1 is assigned to each reduced vertex. I_γ , $I_{\gamma/\gamma_1} \cdots I_{\gamma_c}$ and I_{γ_j} are related by

$$I_\gamma = I_{\gamma/\gamma_1} \cdots I_{\gamma_c} \quad (\text{A.26})$$

Let γ be a subdiagram of $\Gamma \in \mathcal{C}_n$. The dimension $d(\gamma)$ of γ is defined by the dimension of the integral (A.21) which is given by

$$d(\gamma) = 4s(\gamma) + 2v_2(\gamma) - 2l(\gamma) \quad (\text{A.27})$$

Here $s(\gamma)$ is the number of independent internal momenta of γ , $v_2(\gamma)$ the number of 2-vertices, $l(\gamma)$ the number of lines belonging to γ . The dimension $d(\bar{\gamma})$ of the reduced diagram, i.e., the dimension of the corresponding integral (A.24), is given by

$$d(\bar{\gamma}) = 4s(\bar{\gamma}) + 2v_2'(\bar{\gamma}) - 2l(\bar{\gamma}) \quad (\text{A.28})$$

where $v_2'(\bar{\gamma})$ is the number of original 2-vertices belonging to $\bar{\gamma}$ (i.e., not counting those 2-vertices which were obtained by reducing proper self-diagrams of γ). The relation

$$d(\gamma) = d(\bar{\gamma}) + \sum_{a=1}^c d(\gamma_a) \quad (\text{A.29})$$

then follows from (A.27–28). This is obvious in the absence of 2-vertices in γ . Moreover, the insertion of 2-vertices into the lines of γ does not change (A.29) since the degree of a factor $A + B(l^2 - m^2)$ is cancelled by the degree of an additional Feynman denominator. Hence (A.29) holds in general.

We next derive two conditions for the degree $\delta(\gamma)$ defined by (3.15). (A.24) may also be written as

$$d(\gamma) = 4 - r(\gamma) - v_3(\gamma). \quad (\text{A.30})$$

Here $r(\gamma)$ is the number of external lines of γ with those external lines counted twice for which both endpoints lie in γ . $v_3(\gamma)$ denotes the number of 3-vertices in γ . From (3.15) and (A.27) we obtain

$$\delta(\gamma) = d(\gamma) + v_3(\gamma) \quad (\text{A.31})$$

which implies

$$\delta(\gamma) \geq d(\gamma). \quad (\text{A.32})$$

Moreover, (A.29), (A.31) and

$$v_3(\gamma) \geq \sum_{j=1}^c v_3(\gamma_j) \quad (\text{A.33})$$

imply

$$\delta(\gamma) \geq d(\bar{\gamma}) + \sum_{j=1}^c d(\gamma_j). \quad (\text{A.34})$$

The conditions (A.32) and (A.34) guarantee that the final renormalized integral (3.14) is absolutely convergent and approaches a distribution in the limit $\epsilon \rightarrow +0$.⁸

We finally derive the corresponding formulas for the contribution from a diagram $\Delta \in \mathcal{D}_{mn}$ to the Green's function (4.28) of the operator $B_{\{\mu\}}^{(a)}$. Let γ be a subdiagram of Δ . We have to verify that the degree function $\delta_a(\gamma)$ as defined by (4.31) satisfies the conditions

$$\delta_a(\gamma) \geq d(\gamma) \quad (\text{A.35})$$

$$\delta_a(\gamma) \geq d(\bar{\gamma}) + \sum_{j=1}^c \delta_a(\gamma_j) \quad (\text{A.36})$$

for any reduced diagram (A.23). Only the case where γ contains the vertex V_0 need be checked. In this case the dimension of γ is given by

$$d(\gamma) = \sum \#(\mu)_i + 4s(\gamma) + 2v_2(\gamma) - 2l(\gamma) \quad (\text{A.37})$$

which may also be written as

$$d(\gamma) = d - r(\gamma) - v_3(\gamma). \quad (\text{A.38})$$

d is the dimension (4.3) of the operator $B_{\{\mu\}}^{(a)}$, $s(\gamma)$ the number of independent internal momenta of γ , $v_2(\gamma)$ the number of 2-vertices (excluding V_0), $v_3(\gamma)$ the number

⁸ See ref. [5]. Apart from minor changes the convergence proof is not affected by the modifications of this paper.

of 3-vertices (excluding V_0), $l(\gamma)$ the number of lines belonging to γ . (4.30) and (A.38) imply

$$\delta_a(\gamma) = d(\gamma) + v_3(\gamma) + a - d \quad (\text{A.39})$$

yielding (A.35).

In order to check (A.36) we note that (A.37) implies

$$d(\gamma) = d(\bar{\gamma}) + \sum_{j=1}^c d(\gamma_j) \quad (\text{A.40})$$

in the absence of 2-vertices. Since the insertion of a 2-vertex does not change the dimension of a diagram Eq. (A.40) holds in general. Assuming that none of the subdiagrams $\gamma_1, \dots, \gamma_c$ contains V_0 we have

$$\delta_a(\gamma_j) = \delta(\gamma_j). \quad (\text{A.41})$$

Then (A.31), (A.39), (A.41), and

$$v_3(\gamma) \geq \sum_{j=1}^c v_3(\gamma_j) \quad (\text{A.42})$$

imply (A.36). Next we assume that γ_j contains V_0 . Since $\gamma_1, \dots, \gamma_c$ are disjoint only γ_j contains V_0 . Hence

$$\begin{aligned} \delta_a(\gamma_j) &= d(\gamma_j) + v_3(\gamma_j) + a - d \\ \delta_a(\gamma_j) &= d(\gamma_j) + v_3(\gamma_j) \quad \text{if } i \neq j. \end{aligned}$$

Combined with (A.39), (A.42) these relations imply (A.36).

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